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import tensorflow as tf

# Load the MNIST dataset
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data()

# Normalize pixel values to [0, 1]
x_train = x_train.astype('float32') / 255.0
x_test = x_test.astype('float32') / 255.0

# Add a channel dimension to the images
x_train = x_train[..., tf.newaxis]
x_test = x_test[..., tf.newaxis]

# Set up the layers of the network
model = tf.keras.models.Sequential([
    tf.keras.layers.Conv2D(32, (3, 3), activation='relu', input_shape=(28, 28, 1)),
    tf.keras.layers.MaxPooling2D((2, 2)),
    tf.keras.layers.Conv2D(64, (3, 3), activation='relu'),
    tf.keras.layers.MaxPooling2D((2, 2)),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(10, activation='softmax')
])

# Compile the model
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# Train the model on the MNIST dataset
model.fit(x_train, y_train, epochs=3, validation_data=(x_test, y_test))

# Evaluate the model on the MNIST test data
test_loss, test_acc = model.evaluate(x_test, y_test)
print('Test accuracy:', test_acc)


```

 Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz>
 11490434/11490434 ————— 0s 0us/step
 /usr/local/lib/python3.11/dist-packages/keras/src/layers/convolutional/base_conv.py:107: UserWarning: Do not pass an `input\_shape` to `input`
 super().__init__(activity_regularizer=activity_regularizer, **kwargs)
 Epoch 1/3
 1875/1875 ————— 55s 28ms/step - accuracy: 0.8964 - loss: 0.3462 - val_accuracy: 0.9813 - val_loss: 0.0541
 Epoch 2/3
 1875/1875 ————— 52s 28ms/step - accuracy: 0.9836 - loss: 0.0547 - val_accuracy: 0.9872 - val_loss: 0.0378
 Epoch 3/3
 1875/1875 ————— 82s 28ms/step - accuracy: 0.9879 - loss: 0.0402 - val_accuracy: 0.9863 - val_loss: 0.0451
 313/313 ————— 3s 9ms/step - accuracy: 0.9816 - loss: 0.0568
 Test accuracy: 0.986299991607666